Saad Abdullah

Machine Learning Engineer

Curious by nature, always seeking to learn, understand, and improve.

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Education

Erasmus Mundus Joint Master's in Engineering of Data-intensive Intelligent Software Systems (EDISS) 2024–2026
Åbo Akademi University, Finland & Mälardalen University, Sweden | Industrial Machine Learning
Selected as 1 of 27 students from 464 global applicants. Master's thesis (**5/5**): Heterogeneous CPU/GPU execution for sustained YOLO-based mobile inference.

BS Computer Science | FAST-NUCES, Faisalabad 2019–2023
Graduated with a **4.0/4.0 CGPA**, becoming the **second student in campus history** to do so; awarded gold medals throughout the degree; winner of the **7th Programmers' Battle** (C++).
Bachelor's thesis (**4/4**): **Xavier**, an open-source mutation testing toolbox for deep neural networks, built with FastAPI.

Professional Experience

Data & Machine Learning Engineer | Supersight Oy Mar 2025–Present
Helsinki, Finland

- Architected and deployed a production data platform on **GCP**, processing **millions of time-series records daily** using **Delta Lake, Dagster**, and Medallion Architecture.
- Built **Python/Django** analytics and reporting services, transforming large-scale datasets into actionable insights for customer-facing applications.
- Trained and evaluated **YOLO-based computer vision models** on the **LUMI supercomputer**, scaling distributed training to **64 GPUs in parallel** for large-scale person detection.
- Developed experimental **LLM agents using PydanticAI**, integrating tools for automated data analysis, summaries, and visualizations; explored open-source **VLMs** and **LoRA fine-tuning** for synthetic image generation.

Software Engineer | A'sTechware Oct 2020–Jul 2024
Lahore, Pakistan

- Progressed from trainee to senior-level responsibilities, developing and maintaining features across **7+ live, customer-facing web applications**.
- Developed scalable **Ruby on Rails** backends, REST APIs, and integrations for **Word of Mouth**, an Australian review platform serving **millions of users**; integrated **Stripe, HubSpot CRM, Elasticsearch, Redis, and AWS S3**.
- Enhanced **GolfPay360**, a US-based golf course management platform, building features and automating operational workflows using **Ruby on Rails, JavaScript, and scheduled jobs**.
- Modernized legacy Rails applications and improved code quality through **RSpec, Selenium, and Capybara** testing, **Docker**, and **CI/CD**; collaborated directly with clients on requirements and feature delivery.

Projects

Self-Healing Code | VS Code Extension TypeScript, Python, LLMs
Collaborated across **3 European universities** to develop a VS Code extension that detects Python runtime errors, analyzes execution context, and iteratively applies **LLM-generated fixes**, with validation, rollback, and version history. Contributed across architecture, implementation, testing, and distributed code reviews. **[Demo]**

Heterogeneous YOLO Mobile Inference | Master's Thesis YOLO, Android, CPU/GPU
Investigated sustained on-device object detection by partitioning YOLO inference across smartphone CPUs and GPUs. Found that heterogeneous execution benefits depend on **model partitioning and coordination overhead**, rather than parallel execution alone. Thesis awarded **5/5**. **[Demo]**

XRAI | Explainable AI for Medical Image Segmentation Deep Learning, XAI, Computer Vision
Investigated **texture-based learning patterns** in medical image segmentation using saliency maps and explainability techniques, resulting in a **first-author research publication (2026)**.

KHWARIZMI | Multi-Agent AI Analytics Python, FastAPI, LangChain, PostgreSQL
Co-developed a **4-agent AI system** at the **Aalto AI Hackathon 2025**, converting natural-language requests into SQL queries, visualizations, and HTML reports using **OpenAI, Docker, and MinIO**. **[Demo]**